

Raspberry Pi and M2M Technology Power This Smart Street Lighting System



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Venturing into the large-scale application market, a team of five from Bosch has come up with an indigenous smart street lighting solution, powered by Raspberry Pi, Bosch Internet of Things (IoT) Suite and a Zigbee based induction lamp. This innovative solution, Smart Street Lighting Management System (SSLMS), aims to revamp the quality of public lighting by optimising power consumption, and is targeted especially at Indian market.

Reduction in heavy emission and energy consumption can be achieved

For all cities, worldwide, improving efficiency, cost and operation of streetlights is the need of the hour. "It is estimated that as much as 40 per cent of a city's energy budget is spent on outdoor lighting, and that, a single streetlight can emit 200kg of carbon dioxide (CO₂) each year," informs Triveni Prabhu, project manager, Robert Bosch Engineering and Business Solutions Ltd.

The street lighting system
at display during an event



Increasingly, cities need to save energy or do more with less energy, reduce carbon emissions, attain lower operating and maintenance costs, and comply with tighter government regulations. She adds, "Our smart technology enables cities to remotely turn lights on and off, adjust their intensity through automatic programming or sensors, detect energy leakage or possible cable cuts and reduce energy usage."

Two attributes that influenced the production of SSLMS

One fact that drove the team at Bosch to come up with such a solution is that, in European countries, wired communication based brightness control is mostly used, which saves sufficient energy. If brightness of light is reduced by half, the brightness reduction perceived by people is approximately 25 per cent, informs Prabhu. Therefore energy can be saved by reducing brightness.

She adds, "There are also several promising wireless communications currently available that can work for long distance, reliably up to a kilometre, provided line-of-sight is available."

Bosch already had different solutions developed for various applications like Zigbee wireless communication, lighting ballast for brightness control for indoor applications or Web solutions to connect various devices to have various business solutions, notes Prabhu. She says, "These things put together in a street lighting product, in last three to four years of effort from Bosch, has great energy-saving potential."

She adds, "For an energy-hungry country like India, SSLMS is ideal, as it also helps identify and communicate faults to the responsible person, and make payments online, based on the service level.